

Vakil Chapter 25 Note

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1 Cohomology and base change

Let

$$\begin{array}{ccc} X_Z & \xrightarrow{\psi'} & X \\ \pi_Z \downarrow & & \downarrow \pi \\ Z & \xrightarrow{\psi} & Y \end{array}$$

be a Cartesian square, and let $\mathcal{F}$ be a coherent sheaf on X . The natural base-change morphism is

$$\phi_Z^p : \psi^* R^p \pi_* \mathcal{F} \longrightarrow R^p (\pi_Z)_* (\psi'^* \mathcal{F}).$$

For a point $q \in Y$, taking $Z = \operatorname{Spec} k(q)$, we write

$$\phi_q^p : (R^p \pi_* \mathcal{F}) \otimes k(q) \longrightarrow H^p(X_q, \mathcal{F}|_{X_q}).$$

Theorem 1.1 (Cohomology and Base Change, Vakil 25.1.6). *Suppose that $\pi : X \rightarrow Y$ is proper, Y is locally Noetherian, and $\mathcal{F}$ is coherent on X and flat over Y . Fix $q \in Y$ and $p \geq 0$, and assume that ϕ_q^p is surjective. Then:*

- (i) *There is an open neighborhood U of q such that for every morphism $\psi : Z \rightarrow U$, the map ϕ_Z^p is an isomorphism. In particular, ϕ_q^p is an isomorphism.*
- (ii) *The following are equivalent:*
 - (a) ϕ_q^{p-1} is surjective;
 - (b) $R^p \pi_* \mathcal{F}$ is locally free in a neighborhood of q .

With the usual convention in degree -1 , ϕ_q^{-1} is trivially surjective.

Consequently, under these hypotheses,

$$q' \longmapsto h^p(X_{q'}, \mathcal{F}|_{X_{q'}})$$

is locally constant near q precisely in the expected locally free/base-change situation.

Theorem 1.2 (Flat base change, Vakil 24.1.9). *In the Cartesian square above, if ψ is flat (and the usual quasicompact/separated hypotheses needed for higher pushforward hold), then*

$$\psi^* R^p \pi_* \mathcal{F} \xrightarrow{\sim} R^p(\pi_Z)_*(\psi'^* \mathcal{F})$$

for every p .

Remark 1.1. For a finite morphism, the base-change maps are especially simple. This is the first application used in the handwritten notes.

2 Finite flat morphisms: Vakil 25.1.B and 24.3.E

Exercise 2.1 (Vakil 25.1.B / 24.3.E). *Let $\pi : X \rightarrow Y$ be a finite flat morphism of locally Noetherian schemes, and let $\mathcal{F}$ be a finite-rank locally free sheaf on X .*

- (a) *Show that $\pi_* \mathcal{F}$ is a finite-rank locally free sheaf on Y .*
- (b) *If Y is irreducible with generic point η , the degree of π over η is n , and $\mathcal{F}$ has rank r , show that $\pi_* \mathcal{F}$ has rank nr .*

Proof. First check the hypothesis of the Cohomology and Base Change Theorem at every point $q \in Y$.

Because π is finite, it is affine. Locally write

$$Y = \operatorname{Spec} A, \quad X = \operatorname{Spec} B, \quad \mathcal{F} = \widetilde{M}.$$

For $p = 0$, the base-change map is the canonical isomorphism

$$M \otimes_A k(q) \xrightarrow{\sim} M \otimes_B (B \otimes_A k(q)) = H^0(X_q, \mathcal{F}|_{X_q}).$$

For $p > 0$,

$$R^p \pi_* \mathcal{F} = 0$$

because π is affine, and

$$H^p(X_q, \mathcal{F}|_{X_q}) = 0$$

because X_q is affine. Thus every ϕ_q^p is an isomorphism.

Applying Cohomology and Base Change in degree 0, with ϕ_q^{-1} trivially surjective, shows that $\pi_* \mathcal{F}$ is locally free near every q , hence locally free on Y . Finiteness follows from finiteness of π and coherence of $\mathcal{F}$.

For the rank, evaluate at the generic point η :

$$\text{rank}(\pi_* \mathcal{F}) = \dim_{k(\eta)} ((\pi_* \mathcal{F}) \otimes k(\eta)) = \dim_{k(\eta)} H^0(X_\eta, \mathcal{F}|_{X_\eta}).$$

Write

$$X_\eta = \text{Spec } C,$$

where C is a finite-dimensional $k(\eta)$ -algebra of dimension n . Decompose the Artinian ring

$$C \simeq C_1 \times \cdots \times C_s.$$

If

$$H^0(X_\eta, \mathcal{F}|_{X_\eta}) \simeq P_1 \oplus \cdots \oplus P_s,$$

then each P_i is a projective C_i -module of rank r . Since every C_i is Artinian local, P_i is free of rank r . Therefore

$$\dim_{k(\eta)} H^0(X_\eta, \mathcal{F}|_{X_\eta}) = \sum_{i=1}^s r \dim_{k(\eta)} C_i = r \dim_{k(\eta)} C = rn.$$

□

Remark 2.1 (References used in class). For finitely presented modules, flatness is equivalent to projectivity over a local ring; finite projective modules are finite locally free. On an affine scheme $\text{Spec } R$, a coherent sheaf is locally free exactly when its corresponding finite R -module is locally free.

3 Vanishing and base change

Exercise 3.1 (Vakil 25.1.C). Suppose that $\pi : X \rightarrow Y$ is proper, Y is locally Noetherian, and $\mathcal{F}$ is coherent on X and flat over Y . Suppose

$$H^p(X_q, \mathcal{F}|_{X_q}) = 0$$

for some $q \in Y$. Show that there is an open neighborhood U of q such that

$$(R^p \pi_* \mathcal{F})|_U = 0.$$

Proof. The map

$$\phi_q^p : (R^p \pi_* \mathcal{F}) \otimes k(q) \longrightarrow H^p(X_q, \mathcal{F}|_{X_q}) = 0$$

is automatically surjective. By Cohomology and Base Change it is therefore an isomorphism, so

$$(R^p \pi_* \mathcal{F})_q \otimes k(q) = 0.$$

Properness and coherence imply that $R^p \pi_* \mathcal{F}$ is coherent. Nakayama's lemma gives

$$(R^p \pi_* \mathcal{F})_q = 0.$$

Hence the support of $R^p \pi_* \mathcal{F}$ misses some neighborhood U of q , so

$$(R^p \pi_* \mathcal{F})|_U = 0.$$

□

Exercise 3.2 (Vakil 25.1.D). *Under the same hypotheses, suppose*

$$H^p(X_q, \mathcal{F}|_{X_q}) = 0 \quad \text{for every } q \in Y.$$

Show that $(p-1)$ -st cohomology commutes with arbitrary base change: for every $\psi : Z \rightarrow Y$,

$$\phi_Z^{p-1} : \psi^* R^{p-1} \pi_* \mathcal{F} \xrightarrow{\sim} R^{p-1}(\pi_Z)_*(\psi'^* \mathcal{F}).$$

Proof. By Exercise 25.1.C, $R^p \pi_* \mathcal{F}$ vanishes in a neighborhood of every point, hence

$$R^p \pi_* \mathcal{F} = 0.$$

In particular it is locally free. For every $q \in Y$, the map ϕ_q^p is surjective because its target is zero. Part (ii) of the Cohomology and Base Change Theorem therefore implies that

$$\phi_q^{p-1}$$

is surjective for every q .

Part (i) now gives an open neighborhood U_q of every q such that ϕ_Z^{p-1} is an isomorphism for every base change $Z \rightarrow U_q$. The U_q cover Y , and the assertion that a morphism of sheaves is an isomorphism is local on the base. Therefore ϕ_Z^{p-1} is an isomorphism for every $Z \rightarrow Y$. □

Exercise 3.3 (Vakil 25.1.E). *Suppose that $\pi : X \rightarrow Y$ is proper, Y is locally Noetherian, and $\mathcal{F}$ is coherent on X and flat over Y . Suppose*

$$R^p \pi_* \mathcal{F} = 0 \quad \text{for all } p \geq p_0.$$

Show that

$$H^p(X_q, \mathcal{F}|_{X_q}) = 0 \quad \text{for every } q \in Y, \ p \geq p_0.$$

Proof. Fix $q \in Y$. By Grothendieck vanishing,

$$H^N(X_q, \mathcal{F}|_{X_q}) = 0$$

for all sufficiently large N . Choose such an $N \geq p_0$. Then ϕ_q^N is surjective. Since

$$R^N \pi_* \mathcal{F} = 0$$

is locally free, part (ii) of Cohomology and Base Change implies that ϕ_q^{N-1} is surjective. If $N - 1 \geq p_0$, its source is zero, so

$$H^{N-1}(X_q, \mathcal{F}|_{X_q}) = 0.$$

Repeating the same argument descends one degree at a time and gives

$$H^p(X_q, \mathcal{F}|_{X_q}) = 0$$

for every $p \geq p_0$. □

4 Generic base change

Exercise 4.1 (Vakil 25.1.F). *Suppose that $\pi : X \rightarrow Y$ is proper, Y is locally Noetherian and reduced, and $\mathcal{F}$ is coherent on X and flat over Y . Show that there exists a dense open subset $U \subseteq Y$ such that, for every $\psi : Z \rightarrow U$ and every p ,*

$$\phi_Z^p$$

is an isomorphism.

Proof. Let η be a generic point of an irreducible component of Y . Because Y is reduced,

$$\mathcal{O}_{Y,\eta} = k(\eta),$$

so the morphism

$$\mathrm{Spec} k(\eta) \longrightarrow Y$$

is flat. Flat base change therefore gives

$$(R^p \pi_* \mathcal{F}) \otimes k(\eta) \xrightarrow{\sim} H^p(X_\eta, \mathcal{F}|_{X_\eta})$$

for every p . Thus every ϕ_η^p is an isomorphism.

It remains to make the choice of a neighborhood simultaneous in p . Choose an affine neighborhood V of η . Since π is proper, X_V is quasicompact and separated, so choose a finite affine cover

$$X_V = W_1 \cup \cdots \cup W_m.$$

All finite intersections of the W_i are affine. After any affine base change $T \rightarrow V$, these opens and all their intersections remain affine. The resulting Čech complex has length at most m , so for every such T and every $p \geq m$,

$$H^p(X_T, \mathcal{F}_T) = 0.$$

Thus in degrees $p \geq m$ the desired base-change statement is automatic.

For the finitely many degrees $0 \leq p < m$, apply Cohomology and Base Change at η . For each p obtain an open neighborhood $U_{\eta,p} \subseteq V$ such that arbitrary base change is an isomorphism in degree p . Set

$$U_\eta = \bigcap_{p=0}^{m-1} U_{\eta,p}.$$

Then arbitrary base change is an isomorphism in every degree over U_η .

Finally let

$$U = \bigcup_{\eta} U_\eta,$$

where η runs over the generic points of the irreducible components of Y . The set U is open and contains every generic point, hence is dense, and it has the required property. $\square$

Remark 4.1. The finite affine-cover argument is the local uniform bound used in class. It avoids trying to take an infinite intersection of degree-by-degree open sets, and it does not require a global bound on $\dim Y$.

5 Families of curves and Grauert's theorem

Theorem 5.1 (Grauert, Vakil 25.1.5). *Let $\pi : X \rightarrow Y$ be proper, with Y reduced and locally Noetherian, and let $\mathcal{F}$ be coherent on X and flat over Y . If*

$$q \longmapsto h^p(X_q, \mathcal{F}|_{X_q})$$

is locally constant, then $R^p\pi_\mathcal{F}$ is locally free and formation of $R^p\pi_*\mathcal{F}$ commutes with arbitrary base change.*

Exercise 5.2 (Vakil 25.1.G). *Let $\pi : C \rightarrow S$ be a proper smooth morphism whose fibers are smooth curves of genus $g > 1$, with S locally Noetherian.*

(a) *If $n > 1$, show that*

$$\pi_*\Omega_{C/S}^{\otimes n}$$

is a vector bundle of rank

$$(2n-1)(g-1),$$

and that its formation commutes with arbitrary base change $S' \rightarrow S$.

(b) *If S is reduced, show that*

$$\pi_*\Omega_{C/S}$$

is a vector bundle of rank g , and that its formation commutes with arbitrary base change.

Proof. Because π is a smooth family of curves, $\Omega_{C/S}$ is the relative canonical line bundle. For part (a), fix $q \in S$. By Serre duality,

$$H^1\left(C_q, \Omega_{C_q/k(q)}^{\otimes n}\right) \cong H^0\left(C_q, \Omega_{C_q/k(q)}^{\otimes (1-n)}\right)^\vee.$$

Now

$$\deg\left(\Omega_{C_q/k(q)}^{\otimes(1-n)}\right) = (1-n)(2g-2) < 0,$$

so

$$H^1\left(C_q, \Omega_{C_q/k(q)}^{\otimes n}\right) = 0.$$

Exercise 25.1.D, with $p = 1$, implies that H^0 commutes with arbitrary base change. Cohomology and Base Change in degree 0 then shows that $\pi_*\Omega_{C/S}^{\otimes n}$ is locally free.

Its rank is the fiber dimension. Riemann–Roch and the vanishing of H^1 give

$$\begin{aligned} h^0\left(C_q, \Omega_{C_q/k(q)}^{\otimes n}\right) &= \deg\left(\Omega_{C_q/k(q)}^{\otimes n}\right) + 1 - g \\ &= n(2g-2) + 1 - g \\ &= (2n-1)(g-1). \end{aligned}$$

For part (b), Serre duality gives

$$h^0(C_q, \Omega_{C_q/k(q)}) = g$$

for every q . Thus the fiber dimension is constant. Since S is reduced, Grauert's theorem applies and yields that

$$\pi_*\Omega_{C/S}$$

is locally free of rank g , with arbitrary base change. □

Remark 5.1. Part (b) is not obtained from part (a), since part (a) uses $n > 1$. The reduced hypothesis in part (b) can be removed by a different argument, but Grauert gives the stated result immediately under the hypotheses of the exercise.

6 $\mathcal{O}$ -connected morphisms

Definition 6.1. A morphism $\pi : X \rightarrow Y$ is $\mathcal{O}$ -connected if the natural map

$$\mathcal{O}_Y \longrightarrow \pi_*\mathcal{O}_X$$

is an isomorphism.

Exercise 6.2 (Vakil 25.1.I). *Suppose that $\pi : X \rightarrow Y$ is a proper flat morphism of locally Noetherian schemes whose fibers satisfy*

$$h^0(X_q, \mathcal{O}_{X_q}) = 1.$$

Show that π is $\mathcal{O}$ -connected.

Proof. For each $q \in Y$, consider

$$\mathcal{O}_Y \otimes k(q) \longrightarrow (\pi_*\mathcal{O}_X) \otimes k(q) \xrightarrow{\phi_q^0} H^0(X_q, \mathcal{O}_{X_q}).$$

The composition sends 1 to 1. Since the target is a one-dimensional $k(q)$ -vector space, the composition is an isomorphism. In particular ϕ_q^0 is surjective.

By Cohomology and Base Change, ϕ_q^0 is an isomorphism for every q , and $\pi_*\mathcal{O}_X$ is locally free of rank 1. Thus

$$\mathcal{O}_Y \longrightarrow \pi_*\mathcal{O}_X$$

is a morphism of line bundles which is an isomorphism on every fiber. Equivalently, its cokernel has zero fiber at every point; Nakayama's lemma shows that the cokernel is zero. A surjective morphism between line bundles is an isomorphism. Hence

$$\mathcal{O}_Y \xrightarrow{\sim} \pi_*\mathcal{O}_X.$$

□

Remark 6.1. The hypothesis $h^0(X_q, \mathcal{O}_{X_q}) = 1$ holds, in particular, when the fibers are geometrically connected and geometrically reduced.

Exercise 6.3 (Vakil 25.1.J). *Suppose that $\pi : X \rightarrow Y$ is proper and $\mathcal{O}$ -connected. If $\mathcal{M}$ is an invertible sheaf on Y , show that*

$$\mathcal{M} \longrightarrow \pi_*\pi^*\mathcal{M}$$

is an isomorphism. The same argument works for any finite-rank locally free sheaf $\mathcal{M}$.

Proof. By the projection formula,

$$\pi_*\pi^*\mathcal{M} \simeq \pi_*(\mathcal{O}_X \otimes \pi^*\mathcal{M}) \simeq \pi_*\mathcal{O}_X \otimes \mathcal{M} \simeq \mathcal{O}_Y \otimes \mathcal{M} \simeq \mathcal{M}.$$

Under these identifications the displayed isomorphism is the natural map. □

Remark 6.2. The proof of Exercise 25.1.J uses only the condition $\mathcal{O}_Y \simeq \pi_*\mathcal{O}_X$, together with the projection formula. This is the “unit” map

$$\mathcal{M} \longrightarrow \pi_*\pi^*\mathcal{M}.$$

The next proposition concerns the opposite-looking “evaluation” map

$$\pi^*\pi_*\mathcal{L} \longrightarrow \mathcal{L}.$$

Proposition 6.4 (Vakil 25.1.11). *Suppose that $\pi : X \rightarrow Y$ is a flat proper morphism of locally Noetherian schemes with geometrically connected and geometrically reduced fibers. Suppose also that Y is reduced and $\mathcal{L}$ is an invertible sheaf on X whose restriction to every fiber is trivial. Then*

$$\mathcal{M} := \pi_*\mathcal{L}$$

is an invertible sheaf on Y , and the natural evaluation map

$$\pi^*\mathcal{M} \longrightarrow \mathcal{L}$$

is an isomorphism.

Proof. For every $q \in Y$,

$$\mathcal{L}|_{X_q} \simeq \mathcal{O}_{X_q}.$$

The hypotheses on the fibers give

$$h^0(X_q, \mathcal{L}|_{X_q}) = h^0(X_q, \mathcal{O}_{X_q}) = 1.$$

This is constant on Y . Grauert's theorem gives an invertible sheaf

$$\mathcal{M} = \pi_* \mathcal{L}$$

and base-change isomorphisms

$$\mathcal{M} \otimes k(q) \xrightarrow{\sim} H^0(X_q, \mathcal{L}|_{X_q}) \quad (q \in Y).$$

Consider the evaluation map

$$\epsilon : \pi^* \pi_* \mathcal{L} \longrightarrow \mathcal{L}.$$

After restriction to the fiber X_q , it becomes

$$H^0(X_q, \mathcal{L}|_{X_q}) \otimes \mathcal{O}_{X_q} \longrightarrow \mathcal{L}|_{X_q}.$$

Since $\mathcal{L}|_{X_q} \simeq \mathcal{O}_{X_q}$, this map is surjective.

Let $\mathcal{K} = \text{coker}(\epsilon)$. For every point $x \in X$, with image $q = \pi(x)$, the preceding fiberwise surjectivity gives

$$\mathcal{K}_x \otimes k(q) = 0.$$

Hence also

$$\mathcal{K}_x \otimes k(x) = 0,$$

so Nakayama's lemma gives $\mathcal{K}_x = 0$. Thus ϵ is surjective. Both source and target are line bundles, so ϵ is an isomorphism. $\square$

7 Projective bundles

Exercise 7.1 (Vakil 25.1.K). *Let X be a locally Noetherian scheme and let*

$$\text{pr}_1 : X \times \mathbb{P}^n \longrightarrow X$$

be projection onto the first factor. Suppose that $\mathcal{L}$ is a line bundle on $X \times \mathbb{P}^n$ whose degree on every fiber of pr_1 is zero. Show that

$$(\text{pr}_1)_* \mathcal{L}$$

is invertible and that the natural map

$$\text{pr}_1^* (\text{pr}_1)_* \mathcal{L} \longrightarrow \mathcal{L}$$

is an isomorphism.

Proof. For every $q \in X$,

$$\mathcal{L}|_{\{q\} \times \mathbb{P}^n} \simeq \mathcal{O}_{\mathbb{P}_{k(q)}^n}$$

because its degree is zero. Hence

$$H^1(\mathbb{P}_{k(q)}^n, \mathcal{L}|_{\{q\} \times \mathbb{P}^n}) = H^1(\mathbb{P}_{k(q)}^n, \mathcal{O}) = 0.$$

Exercise 25.1.C gives

$$R^1(\mathrm{pr}_1)_* \mathcal{L} = 0,$$

and Exercise 25.1.D gives arbitrary base change in degree 0. Thus

$$(\mathrm{pr}_1)_* \mathcal{L} \otimes k(q) \xrightarrow{\sim} H^0(\mathbb{P}_{k(q)}^n, \mathcal{O}) \simeq k(q).$$

Cohomology and Base Change in degree 0 now shows that $(\mathrm{pr}_1)_* \mathcal{L}$ is locally free of rank 1.

The evaluation map

$$\mathrm{pr}_1^*(\mathrm{pr}_1)_* \mathcal{L} \longrightarrow \mathcal{L}$$

restricts on every fiber to

$$H^0(\mathbb{P}^n, \mathcal{O}) \otimes \mathcal{O} \longrightarrow \mathcal{O},$$

which is surjective. The same Nakayama argument as in Proposition 25.1.11 shows that the evaluation map is an isomorphism. $\square$

Remark 7.1. The proof is the same evaluation-map argument as in Proposition 25.1.11, once local freeness and base change for the pushforward are known. The same method extends to a projective-space bundle $\mathbb{P}_X(\mathcal{E}) \rightarrow X$.

Exercise 7.2 (Vakil 25.1.L). *Let X be a connected Noetherian scheme. Show that*

$$\mathrm{Pic}(X \times \mathbb{P}^n) \simeq \mathrm{Pic}(X) \times \mathbb{Z}.$$

More explicitly, the map

$$\mathrm{Pic}(X) \times \mathbb{Z} \longrightarrow \mathrm{Pic}(X \times \mathbb{P}^n), \quad (\mathcal{A}, d) \longmapsto \mathrm{pr}_1^* \mathcal{A} \otimes \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^n}(d)$$

is an isomorphism.

Proof. Let $\mathcal{M} \in \mathrm{Pic}(X \times \mathbb{P}^n)$. For every $q \in X$, there is a unique integer $d(q)$ such that

$$\mathcal{M}|_{\{q\} \times \mathbb{P}^n} \simeq \mathcal{O}_{\mathbb{P}_{k(q)}^n}(d(q)).$$

We first show that $d(q)$ is locally constant.

Fix $q_0 \in X$, and set $d_0 = d(q_0)$. Choose $N > |d_0|$. Apply upper semicontinuity to

$$\mathcal{M} \otimes \mathrm{pr}_2^* \mathcal{O}(N).$$

After shrinking to a neighborhood U of q_0 ,

$$h^0(\mathbb{P}_{k(q)}^n, \mathcal{O}(d(q) + N)) \leq h^0(\mathbb{P}_{k(q_0)}^n, \mathcal{O}(d_0 + N)) \quad (q \in U).$$

Since $h^0(\mathbb{P}^n, \mathcal{O}(m))$ is strictly increasing for $m \geq 0$, this forces

$$d(q) \leq d_0.$$

Apply the same argument to

$$\mathcal{M}^\vee \otimes \mathrm{pr}_2^* \mathcal{O}(N).$$

After possibly shrinking U again, it gives

$$-d(q) \leq -d_0,$$

hence

$$d(q) \geq d_0.$$

Therefore $d(q) = d_0$ on a neighborhood of q_0 , so $d : X \rightarrow \mathbb{Z}$ is locally constant. Since X is connected, $d(q)$ is constant; call the common value d .

Set

$$\mathcal{N} = \mathcal{M} \otimes \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^n}(-d).$$

Then $\mathcal{N}$ has degree zero on every fiber. By Exercise 25.1.K,

$$\mathcal{A} := (\mathrm{pr}_1)_* \mathcal{N}$$

is an invertible sheaf and

$$\mathrm{pr}_1^* \mathcal{A} \xrightarrow{\sim} \mathcal{N}.$$

Consequently,

$$\mathcal{M} \simeq \mathrm{pr}_1^* \mathcal{A} \otimes \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^n}(d).$$

Thus an inverse to the displayed map in the statement is

$$\mathcal{M} \mapsto ((\mathrm{pr}_1)_*(\mathcal{M} \otimes \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^n}(-d)), d).$$

Hence

$$\mathrm{Pic}(X \times \mathbb{P}^n) \simeq \mathrm{Pic}(X) \times \mathbb{Z}.$$

□

Remark 7.2. The same argument gives the analogous description for a projective-space bundle $\mathbb{P}_X(\mathcal{E}) \rightarrow X$ over a connected Noetherian base, with the relative $\mathcal{O}(1)$ supplying the $\mathbb{Z}$ -factor.

Exercise 7.3 (Vakil 25.1.M). *Suppose that*

$$\pi : X \rightarrow Y$$

is a projective flat morphism over a Noetherian integral scheme, and all geometric fibers are isomorphic to $\mathbb{P}^n$. Show that π is a projective bundle if and only if there is an invertible sheaf $\mathcal{L}$ on X whose restriction to every geometric fiber is $\mathcal{O}(1)$.

Proof. If $X \simeq \mathbb{P}_Y(\mathcal{E})$, the relative $\mathcal{O}_{\mathbb{P}_Y(\mathcal{E})}(1)$ has the required restriction on every geometric fiber.

Conversely, suppose that $\mathcal{L}$ is an invertible sheaf on X such that

$$\mathcal{L}|_{X_{\bar{q}}} \simeq \mathcal{O}_{\frac{\mathbb{P}^n}{k(q)}}(1)$$

for every geometric point $\bar{q} \rightarrow Y$. Then

$$h^0(X_{\bar{q}}, \mathcal{L}|_{X_{\bar{q}}}) = n + 1$$

for all $\bar{q}$. Since Y is integral, hence reduced, Grauert's theorem gives

$$\mathcal{E} := \pi_* \mathcal{L}$$

locally free of rank $n + 1$, and formation of $\mathcal{E}$ commutes with base change.

Consider the evaluation map

$$\pi^* \mathcal{E} \longrightarrow \mathcal{L}.$$

On each geometric fiber it is the standard evaluation map

$$H^0(\mathbb{P}^n, \mathcal{O}(1)) \otimes \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1),$$

which is surjective. By Nakayama's lemma, the global evaluation map is surjective. By the universal property of projectivization, it induces a Y -morphism

$$f : X \longrightarrow \mathbb{P}_Y(\mathcal{E})$$

such that

$$f^* \mathcal{O}_{\mathbb{P}_Y(\mathcal{E})}(1) \simeq \mathcal{L}.$$

After base change to every geometric point $\bar{q}$, the map

$$f_{\bar{q}} : X_{\bar{q}} \longrightarrow \mathbb{P}(\mathcal{E}_{\bar{q}})$$

is the map defined by the complete linear system of $\mathcal{O}(1)$, hence is an isomorphism. Therefore f_q is already an isomorphism for every ordinary point q , by faithfully flat descent.

It remains to show that f itself is an isomorphism. The morphism f is proper, because $X \rightarrow Y$ is proper and $\mathbb{P}_Y(\mathcal{E}) \rightarrow Y$ is separated. Its fibers are finite, so f is quasi-finite. Hence f is finite.

Let $z \in \mathbb{P}_Y(\mathcal{E})$, let q be its image in Y , and let $x \in X$ be the unique point over z . Write

$$A = \mathcal{O}_{\mathbb{P}_Y(\mathcal{E}), z}, \quad B = \mathcal{O}_{X, x}.$$

The finite map gives a finite local homomorphism

$$A \longrightarrow B.$$

Since the fiber map is an isomorphism,

$$A \otimes_{\mathcal{O}_{Y, q}} k(q) \xrightarrow{\sim} B \otimes_{\mathcal{O}_{Y, q}} k(q).$$

The cokernel of $A \rightarrow B$ therefore has zero reduction modulo $\mathfrak{m}_q$, so Nakayama's lemma shows that $A \rightarrow B$ is surjective. Let

$$0 \longrightarrow I \longrightarrow A \longrightarrow B \longrightarrow 0$$

be exact. Because $X \rightarrow Y$ is flat,

$$\mathrm{Tor}_1^{\mathcal{O}_{Y,q}}(k(q), B) = 0.$$

Tensoring the exact sequence with $k(q)$ gives

$$0 \longrightarrow I \otimes k(q) \longrightarrow A \otimes k(q) \longrightarrow B \otimes k(q) \longrightarrow 0.$$

The last map is an isomorphism, so $I \otimes k(q) = 0$. Nakayama's lemma again gives $I = 0$. Thus $A \simeq B$ for every z , and f is an isomorphism:

$$X \simeq \mathbb{P}_Y(\mathcal{E}).$$

□

Remark 7.3 (Class addition). The reducedness used above through Grauert can be avoided. Indeed, on every geometric fiber,

$$H^1(\mathbb{P}^n, \mathcal{O}(1)) = 0.$$

By Cohomology and Base Change,

$$R^1\pi_*\mathcal{L} = 0,$$

the maps ϕ_q^0 are isomorphisms, and $\pi_*\mathcal{L}$ is locally free of rank $n+1$. The rest of the proof is unchanged. Thus this argument does not intrinsically require Y to be reduced.

Remark 7.4 (General projectivization principle). More generally, a surjection

$$\pi^*\mathcal{E} \twoheadrightarrow \mathcal{L},$$

where $\mathcal{E}$ is a vector bundle on Y and $\mathcal{L}$ is a line bundle on X , induces a morphism

$$X \longrightarrow \mathbb{P}_Y(\mathcal{E}).$$

Fiberwise, after choosing a basis of $\mathcal{E}_q$, the map is given by the corresponding homogeneous coordinates. In Exercise 25.1.M, because $\mathcal{E} = \pi_*\mathcal{L}$ and $\mathcal{L}|_{X_{\bar{q}}} \simeq \mathcal{O}(1)$, these fiber maps are the standard isomorphisms onto projective space.

8 Grassmannian “for the third time”

Recall. The Grassmannian functor is

$$G(k, n): \mathbf{Sch}^{\mathrm{op}} \longrightarrow \mathbf{Sets},$$

where

$$G(k, n)(B) = \{ \mathcal{O}_B^{\oplus n} \twoheadrightarrow \mathcal{Q} \mid \mathcal{Q} \text{ is locally free of rank } k \} / \cong.$$

Two quotients $\mathcal{O}_B^{\oplus n} \twoheadrightarrow \mathcal{Q}_1$ and $\mathcal{O}_B^{\oplus n} \twoheadrightarrow \mathcal{Q}_2$ represent the same element if there is an isomorphism $\mathcal{Q}_1 \xrightarrow{\sim} \mathcal{Q}_2$ compatible with the two quotient maps.

Definition 8.1. Define

$$G'(k, n): \mathbf{Sch}^{\text{op}} \longrightarrow \mathbf{Sets}$$

by sending a scheme B to the set of closed immersions

$$i: X \hookrightarrow \mathbf{P}_B^{n-1}$$

such that $X \rightarrow B$ is flat and finitely presented and, for every $b \in B$,

$$X_b \hookrightarrow \mathbf{P}_{\kappa(b)}^{n-1}$$

is a **linear embedding** of $\mathbf{P}_{\kappa(b)}^{k-1}$.

Theorem 8.2 (25.3.4). *The functor $G'(k, n)$ is represented by $G(k, n)$. Equivalently, there is a natural isomorphism of functors*

$$G'(k, n) \xrightarrow{\sim} G(k, n).$$

Proof. We construct natural bijections on B -points and check that the two constructions are inverse to one another.

First recall the easy direction. Given a rank- k locally free quotient

$$\mathcal{O}_B^{\oplus n} \twoheadrightarrow \mathcal{Q},$$

there is an induced closed immersion

$$\text{Proj}_B \text{Sym } \mathcal{Q} \hookrightarrow \text{Proj}_B \text{Sym } \mathcal{O}_B^{\oplus n} = \mathbf{P}_B^{n-1}.$$

Thus we obtain a natural map

$$G(k, n)(B) \longrightarrow G'(k, n)(B).$$

Remark 8.1. Affine-locally, let $V = \text{Spec } A \subseteq B$ and trivialize $\mathcal{Q}|_V \simeq \mathcal{O}_V^{\oplus k}$. If

$$e_i \longmapsto \sum_{j=1}^k a_{ij} f_j,$$

then the induced closed immersion $\mathbf{P}_A^{k-1} \hookrightarrow \mathbf{P}_A^{n-1}$ comes from the graded surjection

$$A[x_1, \dots, x_n] \longrightarrow A[y_1, \dots, y_k], \quad x_i \longmapsto \sum_{j=1}^k a_{ij} y_j.$$

The opens

$$D_+ \left(\sum_j a_{ij} y_j \right)$$

cover $\mathbf{P}_A^{k-1}$. Over a field this is the usual formula

$$[c_1 : \dots : c_k] \longmapsto \left[\sum_j a_{1j} c_j : \dots : \sum_j a_{kj} c_j \right].$$

Now construct the other direction

$$G'(k, n)(B) \longrightarrow G(k, n)(B).$$

This is where cohomology and base change enter.

Let

$$\begin{array}{ccc} X & \xhookrightarrow{i} & \mathbf{P}_B^{n-1} \\ & \searrow f & \downarrow \pi \\ & & B \end{array}$$

represent an element of $G'(k, n)(B)$. From

$$0 \longrightarrow \mathcal{I}_X \longrightarrow \mathcal{O}_{\mathbf{P}_B^{n-1}} \longrightarrow i_* \mathcal{O}_X \longrightarrow 0$$

we obtain, after twisting by $\mathcal{O}(1)$,

$$0 \longrightarrow \mathcal{I}_X(1) \longrightarrow \mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \longrightarrow i_* \mathcal{O}_X \otimes \mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \longrightarrow 0.$$

For $b \in B$, consider the Cartesian square

$$\begin{array}{ccc} X_b & \xhookrightarrow{i_b} & \mathbf{P}_{\kappa(b)}^{n-1} \\ \downarrow & & \downarrow \\ X & \xhookrightarrow{i} & \mathbf{P}_B^{n-1}. \end{array}$$

Because i is a closed immersion,

$$\left(i_* \mathcal{O}_X \otimes \mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \right) \big|_{\mathbf{P}_{\kappa(b)}^{n-1}} \simeq (i_b)_* \mathcal{O}_{X_b}(1).$$

Hence

$$\begin{aligned} H^j \left(\mathbf{P}_{\kappa(b)}^{n-1}, (i_* \mathcal{O}_X \otimes \mathcal{O}(1)) \big|_{\mathbf{P}_{\kappa(b)}^{n-1}} \right) &\simeq H^j(X_b, \mathcal{O}_{X_b}(1)) \\ &\simeq H^j(\mathbf{P}_{\kappa(b)}^{k-1}, \mathcal{O}(1)). \end{aligned}$$

Therefore

$$H^j(X_b, \mathcal{O}_{X_b}(1)) = 0 \quad (j > 0),$$

and

$$h^0(X_b, \mathcal{O}_{X_b}(1)) = k.$$

By cohomology and base change,

$$\mathcal{E} := \pi_* \left(i_* \mathcal{O}_X \otimes \mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \right)$$

is locally free of rank k , and formation of $\mathcal{E}$ commutes with base change.

Applying π_* to the twisted exact sequence gives

$$0 \longrightarrow \pi_* \mathcal{I}_X(1) \longrightarrow \pi_* \mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \longrightarrow \mathcal{E} \longrightarrow R^1 \pi_* \mathcal{I}_X(1) \longrightarrow \cdots$$

We claim that

$$\pi_* \mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \longrightarrow \mathcal{E}$$

is surjective. For every $b \in B$, the base-change square is

$$\begin{array}{ccc} \pi_* \mathcal{O}(1) \otimes \kappa(b) & \longrightarrow & \mathcal{E} \otimes \kappa(b) \\ \sim \downarrow & & \downarrow \sim \\ H^0(\mathbf{P}_{\kappa(b)}^{n-1}, \mathcal{O}(1)) & \longrightarrow & H^0(X_b, \mathcal{O}_{X_b}(1)). \end{array}$$

The lower horizontal map is surjective because $X_b \hookrightarrow \mathbf{P}_{\kappa(b)}^{n-1}$ is a linear embedding. Thus the cokernel of the upper map has zero fibre at every b , and Nakayama's lemma gives the claim.

Since

$$\pi_* \mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \simeq \mathcal{O}_B^{\oplus n},$$

we obtain a rank- k locally free quotient

$$\mathcal{O}_B^{\oplus n} \twoheadrightarrow \mathcal{E}.$$

This defines

$$G'(k, n)(B) \longrightarrow G(k, n)(B).$$

The construction is functorial in B because of base change.

It remains to check that the two constructions are inverse.

Start with a quotient

$$\mathcal{O}_B^{\oplus n} \twoheadrightarrow \mathcal{Q}, \quad \text{rk } \mathcal{Q} = k,$$

and let

$$X = \text{Proj}_B \text{Sym } \mathcal{Q} \hookrightarrow \mathbf{P}_B^{n-1}.$$

Writing $f: X \rightarrow B$, the quotient reconstructed from X is

$$\mathcal{O}_B^{\oplus n} \twoheadrightarrow \pi_* \left(\mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \otimes i_* \mathcal{O}_X \right).$$

By the projection formula,

$$\begin{aligned} \pi_* \left(\mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \otimes i_* \mathcal{O}_X \right) &\simeq \pi_* i_* i^* \mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \\ &= f_* \mathcal{O}_X(1) \\ &\simeq \mathcal{Q}. \end{aligned}$$

Thus

$$G(k, n) \longrightarrow G'(k, n) \longrightarrow G(k, n)$$

is the identity.

Conversely, start with

$$X \hookrightarrow \mathbf{P}_B^{n-1}$$

in $G'(k, n)(B)$, and put

$$\mathcal{E} = \pi_* \left(\mathcal{O}_{\mathbf{P}_B^{n-1}}(1) \otimes i_* \mathcal{O}_X \right).$$

The quotient

$$\mathcal{O}_B^{\oplus n} \twoheadrightarrow \mathcal{E}$$

produces

$$\mathrm{Proj}_B \mathrm{Sym} \mathcal{E} \hookrightarrow \mathbf{P}_B^{n-1}.$$

By the relative projective-bundle construction recalled in §25.1M, the evaluation map gives a natural isomorphism

$$X \xrightarrow{\sim} \mathrm{Proj}_B \mathrm{Sym} \mathcal{E}$$

compatible with the embeddings into $\mathbf{P}_B^{n-1}$. Hence

$$G'(k, n) \longrightarrow G(k, n) \longrightarrow G'(k, n)$$

is also the identity.

Thus $G(k, n)$ and $G'(k, n)$ are the same moduli functor. □

9 Hypersurfaces

Definition 9.1. Define

$$H_{d,n}: \mathbf{Sch}^{\mathrm{op}} \longrightarrow \mathbf{Sets}$$

by letting $H_{d,n}(B)$ consist of closed immersions

$$X \hookrightarrow \mathbf{P}_B^n$$

such that $X \rightarrow B$ is flat and finitely presented and, for every $b \in B$,

$$X_b \hookrightarrow \mathbf{P}_{\kappa(b)}^n$$

is a degree- d hypersurface.

Put

$$M = \binom{n+d}{d}, \quad N = M - 1 = \binom{n+d}{d} - 1.$$

Proposition 9.2 (25.3.6). *The functor $H_{d,n}$ is represented by*

$$\mathbf{P}^N = \mathbf{P}^{\binom{n+d}{d}-1}.$$

Proof. As in the Grassmannian case, one direction comes from the universal family, while **the other direction requires cohomology and base change**.

Step 1: the universal hypersurface. Use homogeneous coordinates $[a_I]_{|I|=d}$ on $\mathbf{P}^N$ and $[x_0 : \cdots : x_n]$ on $\mathbf{P}^n$. On $\mathbf{P}^n \times \mathbf{P}^N$, consider the universal bihomogeneous form

$$F = \sum_{|I|=d} a_I x^I,$$

which is a section of $\mathcal{O}(d, 1)$. Let

$$\mathcal{X} = V(F) \hookrightarrow \mathbf{P}^n \times \mathbf{P}^N.$$

The fibre over $s = [a_I] \in \mathbf{P}^N$ is

$$\mathcal{X}_s = V\left(\sum_{|I|=d} a_I(s)x^I\right) \subseteq \mathbf{P}_{\kappa(s)}^n.$$

This is the natural geometric picture: a point of the parameter space is the coefficient vector of a degree- d homogeneous equation.

The morphism $\mathcal{X} \rightarrow \mathbf{P}^N$ is finitely presented. For flatness we use the following result that was marked in the notes.

Theorem 9.3 (24.6.9: slicing criterion for flatness on the source). *Suppose $(B, \mathfrak{n}) \rightarrow (A, \mathfrak{m})$ is a local morphism of Noetherian local rings, M_0 is a finitely generated A -module, and $t \in \mathfrak{m}$ is a nonzerodivisor on $M_0/\mathfrak{n}M_0$. If M_0 is flat over B , then M_0/tM_0 is flat over B .*

Fix $x \in \mathcal{X}$, let $s \in \mathbf{P}^N$ be its image, and set

$$B_s = \mathcal{O}_{\mathbf{P}^N, s}, \quad A_x = \mathcal{O}_{\mathbf{P}^n \times \mathbf{P}^N, x}.$$

The ring A_x is flat over B_s , because $\mathbf{P}^n \times \mathbf{P}^N \rightarrow \mathbf{P}^N$ is flat. Let $t \in A_x$ be the local equation induced by F . Modulo the maximal ideal of B_s , t becomes the specialized homogeneous polynomial

$$F_s = \sum_{|I|=d} a_I(s)x^I$$

in the local ring $\mathcal{O}_{\mathbf{P}_{\kappa(s)}^n, x}$. At least one coefficient $a_I(s)$ is nonzero, because s is a point of projective space. Hence $F_s \neq 0$; since the local ring of $\mathbf{P}_{\kappa(s)}^n$ is a domain, F_s is a nonzerodivisor. The slicing criterion therefore shows that

$$\mathcal{O}_{\mathcal{X}, x} = A_x/(t)$$

is flat over B_s . Thus $\mathcal{X} \rightarrow \mathbf{P}^N$ is flat and finitely presented.

Consequently, every morphism $f: B \rightarrow \mathbf{P}^N$ yields, by base change, a family

$$\mathcal{X}_f = \mathcal{X} \times_{\mathbf{P}^N} B \hookrightarrow \mathbf{P}_B^n,$$

and therefore a natural map

$$\mathrm{Hom}_{\mathbf{Sch}}(B, \mathbf{P}^N) \longrightarrow H_{d,n}(B).$$

Step 2: from a family of hypersurfaces to a map to $\mathbf{P}^N$.

Let

$$\begin{array}{ccc} X & \xhookrightarrow{i} & \mathbf{P}_B^n \\ & \searrow f & \downarrow \pi \\ & & B \end{array}$$

represent an element of $H_{d,n}(B)$. We have

$$0 \longrightarrow \mathcal{I}_X \longrightarrow \mathcal{O}_{\mathbf{P}_B^n} \longrightarrow i_*\mathcal{O}_X \longrightarrow 0,$$

and after twisting by d ,

$$0 \longrightarrow \mathcal{I}_X(d) \longrightarrow \mathcal{O}_{\mathbf{P}_B^n}(d) \longrightarrow i_*\mathcal{O}_X(d) \longrightarrow 0.$$

For each $b \in B$, the fibre $X_b \subseteq \mathbf{P}_{\kappa(b)}^n$ is a degree- d hypersurface, so

$$\mathcal{I}_{X_b}(d) \simeq \mathcal{O}_{\mathbf{P}_{\kappa(b)}^n}.$$

Thus

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}_{\kappa(b)}^n} \longrightarrow \mathcal{O}_{\mathbf{P}_{\kappa(b)}^n}(d) \longrightarrow \mathcal{O}_{X_b}(d) \longrightarrow 0.$$

It follows that

$$H^j(X_b, \mathcal{O}_{X_b}(d)) = 0 \quad (j > 0)$$

and

$$h^0(X_b, \mathcal{O}_{X_b}(d)) = \binom{n+d}{d} - 1 = N.$$

Therefore cohomology and base change gives that

$$\mathcal{E} := \pi_* i_* \mathcal{O}_X(d)$$

is locally free of rank N , with formation commuting with base change.

As before, the map

$$\pi_* \mathcal{O}_{\mathbf{P}_B^n}(d) \longrightarrow \mathcal{E}$$

is surjective. Indeed, after tensoring with $\kappa(b)$, it becomes

$$H^0(\mathbf{P}_{\kappa(b)}^n, \mathcal{O}(d)) \twoheadrightarrow H^0(X_b, \mathcal{O}_{X_b}(d)),$$

which is surjective by the fibre exact sequence above; Nakayama's lemma then gives global surjectivity.

Since

$$\pi_* \mathcal{O}_{\mathbf{P}_B^n}(d) \simeq \mathcal{O}_B^{\oplus M},$$

we obtain an exact sequence

$$0 \longrightarrow \pi_* \mathcal{I}_X(d) \longrightarrow \mathcal{O}_B^{\oplus M} \longrightarrow \mathcal{E} \longrightarrow 0.$$

Hence

$$\mathcal{L}_0 := \pi_* \mathcal{I}_X(d)$$

is locally free of rank 1. Dualizing gives a surjection

$$\mathcal{O}_B^{\oplus M} \twoheadrightarrow \mathcal{L}_0^\vee.$$

The notes invoke the following standard description of projective space.

Theorem 9.4 (15.2.2). *For a fixed scheme Y , morphisms $Y \rightarrow \mathbf{P}^N$ are in bijection with data*

$$(\mathcal{L}, s_0, \dots, s_N),$$

where $\mathcal{L}$ is an invertible sheaf and $s_0, \dots, s_N \in H^0(Y, \mathcal{L})$ have no common zero, up to isomorphism of these data.

Applying this to the quotient

$$\mathcal{O}_B^{\oplus M} \twoheadrightarrow \mathcal{L}_0^\vee$$

produces a morphism

$$B \longrightarrow \mathbf{P}^N.$$

Thus we have constructed

$$H_{d,n}(B) \longrightarrow \mathrm{Hom}_{\mathrm{Sch}}(B, \mathbf{P}^N).$$

Step 3: the two constructions are inverse.

First start with

$$f: B \longrightarrow \mathbf{P}^N$$

and write

$$\mathcal{L} = f^* \mathcal{O}_{\mathbf{P}^N}(1).$$

The universal hypersurface has ideal sheaf

$$\mathcal{I}_{\mathcal{X}/(\mathbf{P}^n \times \mathbf{P}^N)} \simeq \mathcal{O}_{\mathbf{P}^n \times \mathbf{P}^N}(-d, -1).$$

Because $\mathcal{X} \rightarrow \mathbf{P}^N$ is flat, Exercise 24.2H identifies the ideal sheaf of the pullback with the pullback of the ideal sheaf. Hence

$$\mathcal{I}_{\mathcal{X}_f/\mathbf{P}_B^n} \simeq \mathcal{O}_{\mathbf{P}_B^n}(-d) \otimes \pi^* \mathcal{L}^\vee.$$

After twisting by d and pushing forward,

$$\pi_* \mathcal{I}_{\mathcal{X}_f/\mathbf{P}_B^n}(d) \simeq \pi_* \pi^* \mathcal{L}^\vee \simeq \mathcal{L}^\vee.$$

Therefore the reconstructed quotient is

$$\mathcal{O}_B^{\oplus M} \twoheadrightarrow \mathcal{L},$$

with the same generating sections obtained by pulling back the standard coordinate sections of $\mathcal{O}_{\mathbf{P}^N}(1)$. Thus the reconstructed morphism $B \rightarrow \mathbf{P}^N$ is exactly f . So

$$\mathrm{Hom}(B, \mathbf{P}^N) \longrightarrow H_{d,n}(B) \longrightarrow \mathrm{Hom}(B, \mathbf{P}^N)$$

is the identity.

Conversely, start with

$$X \hookrightarrow \mathbf{P}_B^n$$

in $H_{d,n}(B)$, and put

$$\mathcal{L}_0 = \pi_* \mathcal{I}_X(d).$$

Let

$$f: B \longrightarrow \mathbf{P}^N$$

be the morphism constructed from

$$\mathcal{O}_B^{\oplus M} \twoheadrightarrow \mathcal{L}_0^\vee.$$

Then

$$f^* \mathcal{O}_{\mathbf{P}^N}(-1) \simeq \mathcal{L}_0.$$

If

$$X' = \mathcal{X} \times_{\mathbf{P}^N} B \hookrightarrow \mathbf{P}_B^n,$$

then, again by Exercise 24.2H,

$$\mathcal{I}_{X'/\mathbf{P}_B^n} \simeq \mathcal{O}_{\mathbf{P}_B^n}(-d) \otimes \pi^* \mathcal{L}_0.$$

There is a natural evaluation map

$$\pi^* \pi_* \mathcal{I}_X(d) \longrightarrow \mathcal{I}_X(d).$$

For every $b \in B$, flat pullback of ideal sheaves gives

$$\mathcal{I}_X(d)|_{\mathbf{P}_{\kappa(b)}^n} \simeq \mathcal{I}_{X_b}(d) \simeq \mathcal{O}_{\mathbf{P}_{\kappa(b)}^n}.$$

by previous conclusions (using Cohomology and Base Change Theorem and Nakayama), the evaluation map is indeed an isomorphism:

$$\pi^* \mathcal{L}_0 \xrightarrow{\sim} \mathcal{I}_X(d).$$

Consequently,

$$\mathcal{I}_{X'/\mathbf{P}_B^n} \simeq \mathcal{O}_{\mathbf{P}_B^n}(-d) \otimes \pi^* \mathcal{L}_0 \simeq \mathcal{I}_X.$$

Hence $X' = X$ as closed subschemes of $\mathbf{P}_B^n$, so

$$H_{d,n}(B) \longrightarrow \mathrm{Hom}(B, \mathbf{P}^N) \longrightarrow H_{d,n}(B)$$

is the identity.

Therefore

$$H_{d,n} \simeq \mathrm{Hom}_{\mathbf{Sch}}(-, \mathbf{P}^N),$$

which proves that $H_{d,n}$ is represented by $\mathbf{P}^{\binom{n+d}{d}-1}$. □